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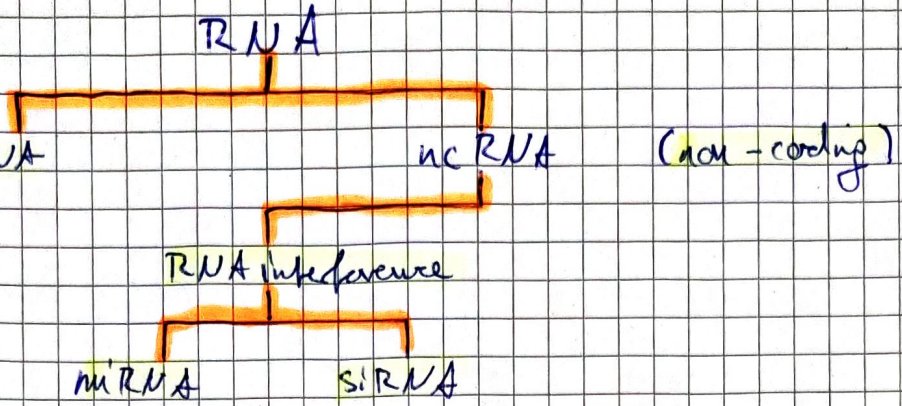
Non-Coding RNAs

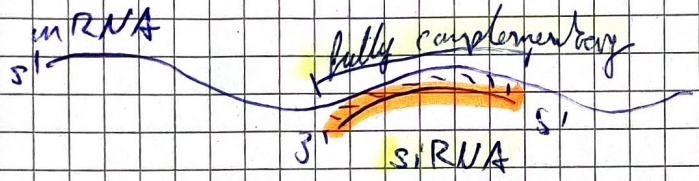
How can we explain differences between cell types of tissues?



RNA Interference (RNAi)

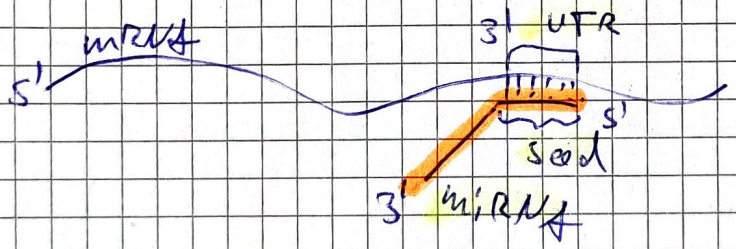
→ a biological process in which RNA molecules are involved in sequence specific suppression of gene expression

a)



also known as post-transcriptional gene silencing

b)



MicroRNA (miRNA) Gene Regulation

→ 19 - 22 nucleotides long

→ key regulators of gene expression

→ each miRNA can potentially regulate hundreds of genes

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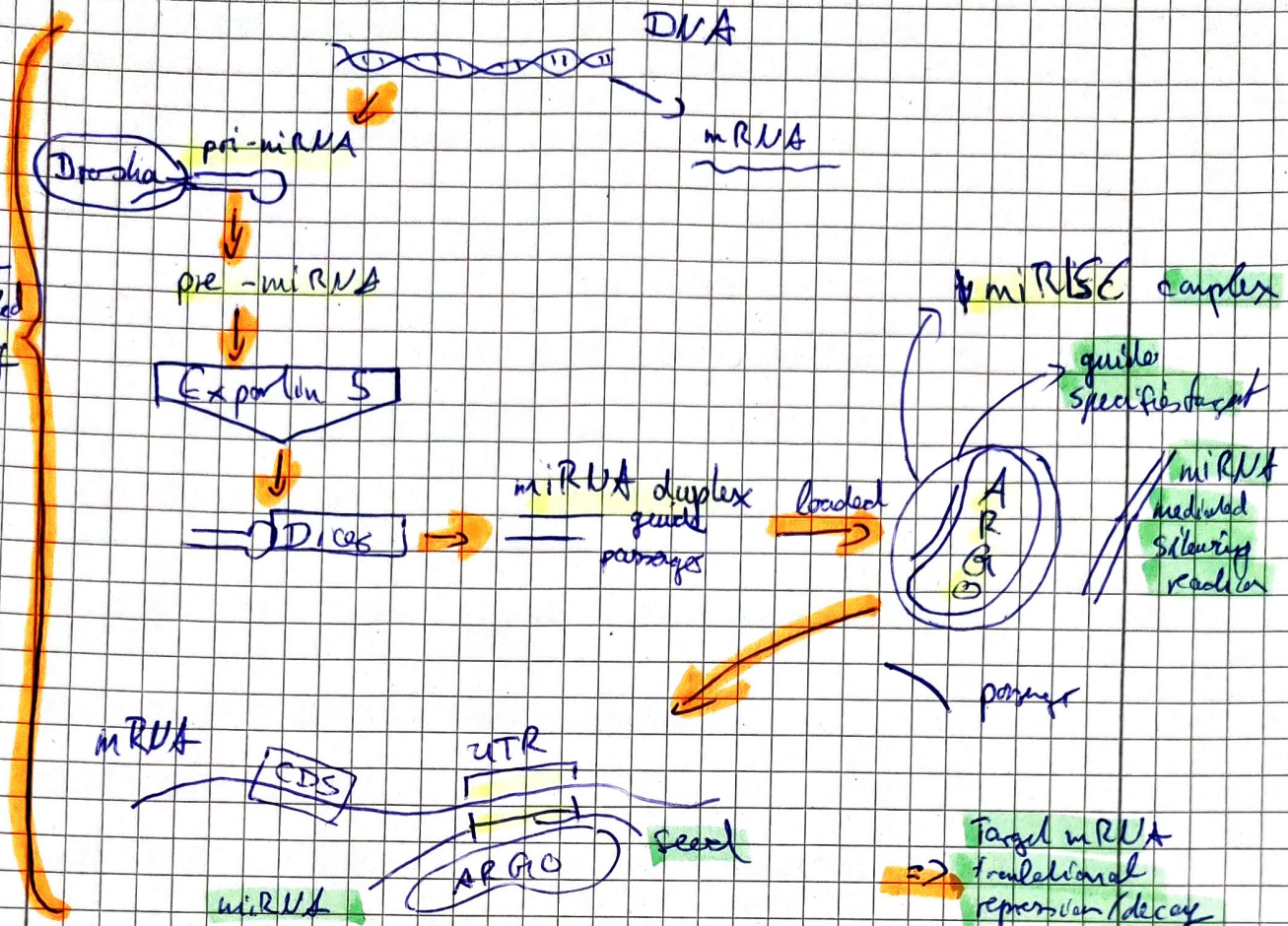
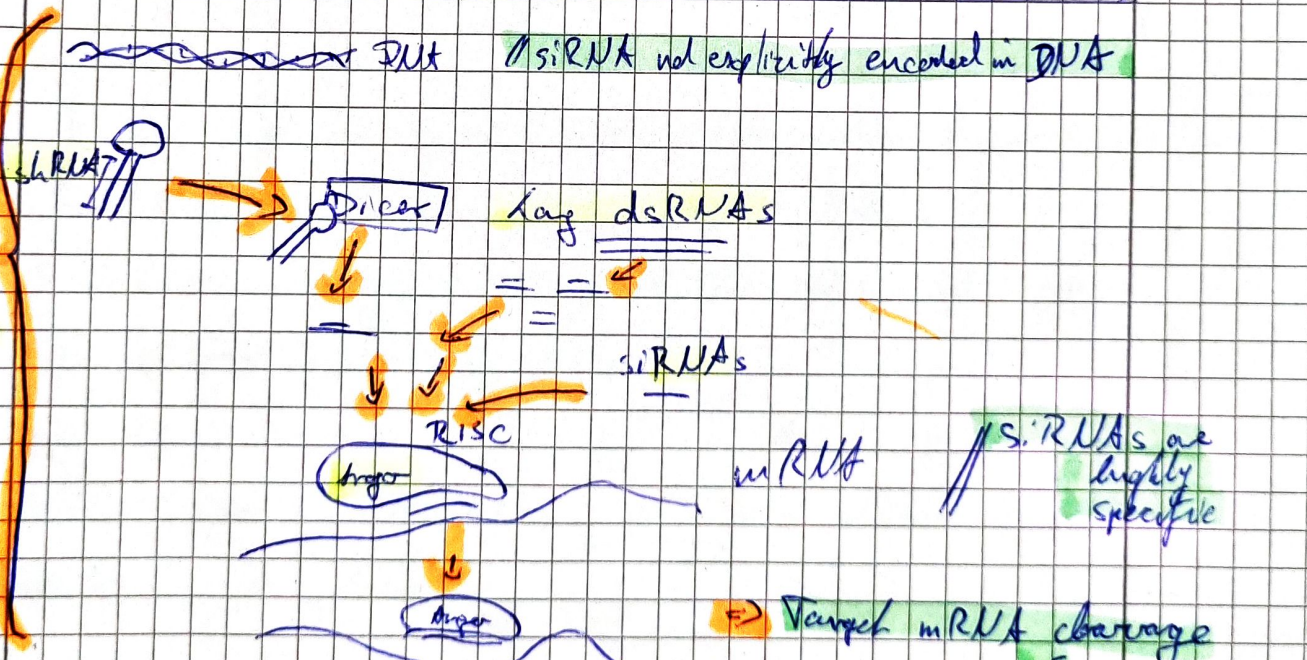
Origin of miRNA





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RNA Interference

→ gene silencing on a genome wide scale

Robotic high-throughput screens

Bioinformatics Tasks:
- Quality Control
- Normalization
- Batch effect correction
- Visual Analytics
- Statistical Analysis
- Systems Biology
- Network Enrichment

RNAi for Therapy

→ RNAi has great potential for targeted therapy

→ But CRISPR is better nowadays

CRISPR - Cas 9

CRISPR = Clusters of regularly interspaced short palindromic repeats

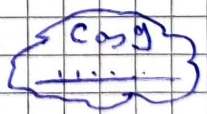
→ natural defense mechanism against viruses found in bacteria

→ Adaptive Immunity

Bacteria incorporate parts of foreign DNA into their genome cut

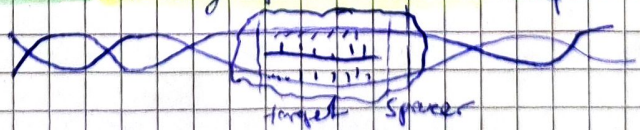
use it for target recognition upon subsequent infection

→ Cleavage mechanism of RNA-guided Cas 9 cut predicted its potential use in programmed genome editing



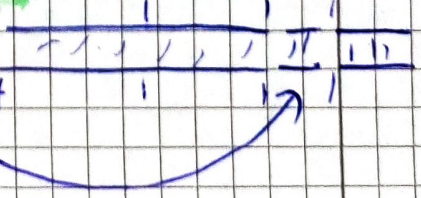
① Cas 9 protein forms a complex with guide RNA in cell

② This complex attaches to a matching genomic DNA sequence adjacent to a spacer



③ Cas 9 - RNA complex cuts the double strands of the DNA

④ Programmed DNA may be inserted at the cut



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CRISPR-Cas 9 vs RNAi

CRISPR-Cas 9	RNAi
→ Permanent gene disruption (knockout)	→ Temporary gene disruption (knockdown)
→ Can not be used for study essential genes	→ Can be used to study essential genes
→ Precise gene targeting with fewer off targets	→ More off target events Silencing multiple transcripts
→ Lower risk of immune response	→ High risk of immune response

miRNA in disease

→ miRNA play a major role in diseases and are increasingly recognized as biomarkers and therapeutic targets

↳ How to identify target genes?

↳ computational prediction needed eg: miRanda

→ miRNA seed pairing
→ Conservation of miRNA binding sites
→ Hybridization energy
→ Accessibility of binding sites

Tissue Specificity

→ miRNA's are short!

→ sequence-based predictions lead to high false positive rate
→ Main Reason: miRNA and target gene may not be expressed at the same time in the same place
→ Tissue-Specific expression data can be used to remove FP's

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Leverage Experimental Data for predictions

Method	Specificity	Sensitivity
CLIP-Seq	good	bad
Computational Prediction	bad	good

↓ many FP

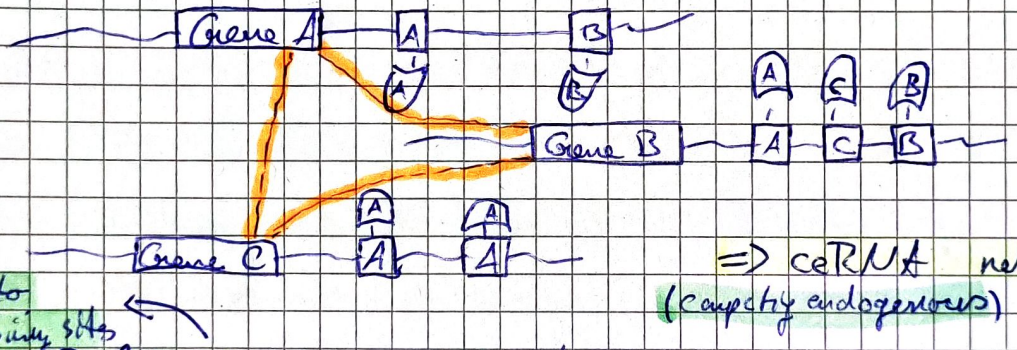
→ use eg's found in experimental data in ML to learn patterns of 'good' binding sites and allow for better prediction of binding sites not seen in experiments

→ allow for binding sites also in coding-region of a gene

PAR-clip data ⇒ Feature Extraction & Analysis ⇒ Feature Selection ⇒ Scoring ⇒ Coding CDS & 3'UTR targets ⇒ Independent testing on htpd

Competing endogenous RNAs

□ miRNA binding site
miRNA



⇒ ceRNA network (compete endogenously)

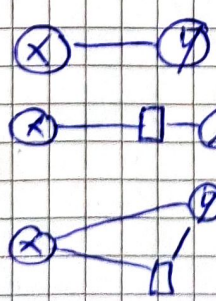
Hard to detect binding sites

→ Problem: miRNA binding site for multiple genes (can be detected using CMI or sensitivity correlation)

→ those ceRNA can regulate each other (cross-regulation)

↳ share competition in binding same sequences

⇒ ceRNA models: ○ gene □ miRNA



gene-gene correlation

gene-miRNA interactions only

gene gene miRNA triplets

→ Tool SPONGE: Multiple Sensitivity Correlation: Significance analysis, Runtime & Multiple # miRNAs

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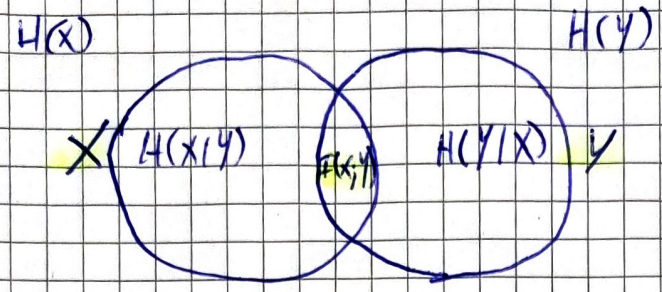
Mutual Information

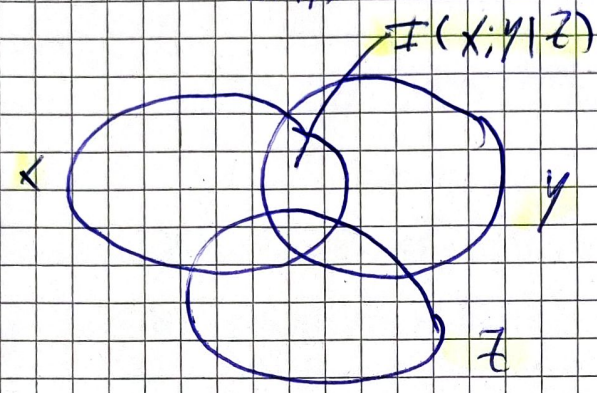
-> Amount of information obtained about one random variable by observing the other random variable

a)



Mutual Information

b)



Conditional mutual information

Issues with Conditional Mutual Information for cERNI network inferences?

-> CMI can be computed for one triplet at a time

-> p-Values are combined with Fishers Method, leading to a loss of power

-> random permutations of gene expression vectors needed to infer an empirical p-Value

-> no genome-wide inference

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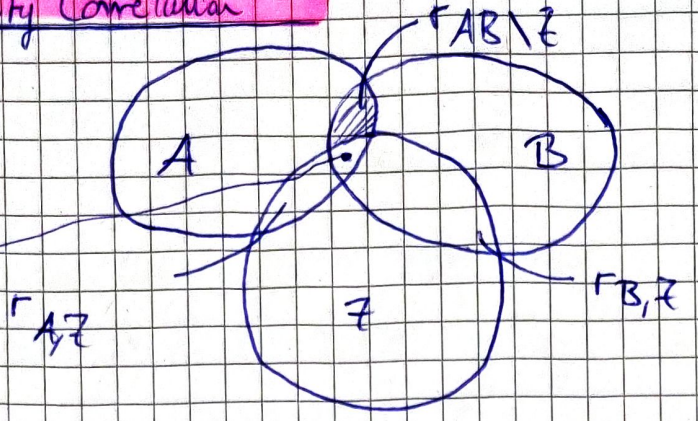
Sensitivity Correlation



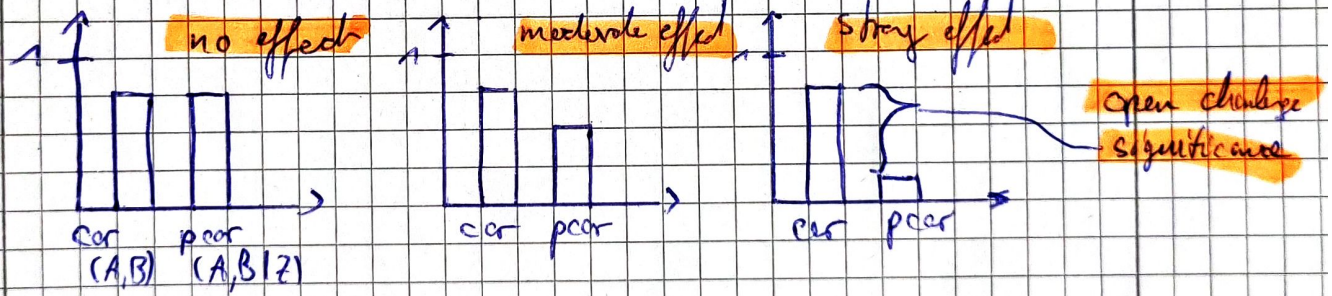
Partial correlation (pcor):

$$r_{AB\setminus Z} = \frac{r_{AB} - (r_{AZ})(r_{BZ})}{\sqrt{1-r_{AZ}^2}\sqrt{1-r_{BZ}^2}}$$

- We are interested in quantifying the regulatory effect of the miRNA Z on genes A and B

Sensitivity Correlation:

$$SCor_{AB|Z} = r_{AB} - r_{AB\setminus Z} = r_{AB} - pcor$$



Multiple sensitivity correlation:

$$mscor(g_1, g_2, M) = cor(g_1, g_2) - pcor(g_1, g_2 | M)$$

Null Model for Scor Values

Null hypothesis: Z has no influence on A and B, thus scor = 0

It follows that $pcor(A,B|Z) = cor(A,B)$

We randomly sample r_{AB} & r_{AZ}. r_{ABZ} can be computed by solving a quadratic formula: $r_{BZ} = \frac{-B \pm \sqrt{AC}}{2A}$

↳ extend idea for multiple miRNAs

↳ Correlation Matrix: $R = \begin{pmatrix} R_{11} & R_{12} \\ R_{21} & R_{22} \end{pmatrix}$

↳ goal: valid eigs of correlation matrices for which m.scor = 0 holds

↳ Simulate Data under null hypothesis

↳ Multiple miRNA sensitivity correlation is affected by

- -> gene-gene correlation (R)
- -> number of miRNAs (m)
- -> # samples

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SPONGE → Infer genome-wide ceRNA interaction Networks

Sparse Partial correlation ON Gene Expression

Workflow:

Gene Expression ↓ | miRNA expression ↓ | Target Scan miR code ↙

1) Gene-miRNA seed match and regression filter // > threshold
   Tissue and cell-type specificity expression based filtering
   ↳ reduces FP's
2) Compute mscor ceRNAs with shared miRNAs
3) Null model based significance analysis
4) Network construction

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① MicroRNAs repress gene expression,
find their target by binding in a seed match region,
act in a complex with an Argonaute protein,
differ from siRNAs which show perfect complementarity and high specificity.
Target prediction suffers from false positives

② miRBase

miRNA family: miRNAs with identical seed sequence

③ Competing endogenous RNAs

→ miRNA binding sites are called: miRNA response elements
→ ~~non~~ According to ceRNA hypothesis, miRNA act in trans

④ SPONGE

How does SPONGE identify gene-miRNA interactions?
→ by using user-provided miRNA target annotation, eg: TargetScan
→ by computing an elastic net regression model with the gene as response variable and the miRNA expression as explanatory variable

How does SPONGE assign p-values to interactions?
→ it uses a null model in which mscor values are simulated under the null hypothesis that the partial correlation equals the correlation of the genes

CSB - L10

Network Alignment

Biological Networks: model entities and their relationships

eg: protein - protein interactions

Network Alignment:

Goal: Finding the best mapping between nodes from two or more networks

Aims is to find topologically or functionally similar nodes or regions in the networks

eg: commonly applied to PPI networks of different species: identify orthologous genes.

(& transfer knowledge from well-studied models)

→ **Local Alignment:** Many-to-many mapping of small subnetworks

→ **Global Alignment:** One-to-one mapping of entire networks

MCES Problem

Maximum common edge subgraph problem:

Find the pairwise, global alignment with the most edges shared between the networks.

$G_1 = (V_1, E_1)$, $G_2 = (V_2, E_2)$, $|V_1| \leq |V_2|$

mapping: $f: V_1 \to V_2$

that maximizes the fitness function $F(f)$:

$$F(f) = \sum_{(u,v) \in E_1} \begin{cases} 1 & , \text{if } (f(u), f(v)) \in E_2 \\ 0 & , \text{else} \end{cases}$$

NP-Hard

CSB

Complexity

⇒ Two networks G1 and G2 are isomorphic if there exists a function $f: V_1 \to V_2$

$\forall_{xy}$ Edges $(x,y) \in E_1 \rightarrow \forall$ Edges $(f(x), f(y)) \in E_2$

⇒ MCES is a generalization of the subgraph isomorphism problem

⇒ Two networks G1 & G2 are isomorphic, iff a mapping f exists preserving all edges $F(f) = |E_1| = |E_2|$

CytoMCS

Common subgraph detection Tool

⇒ iterative local search algorithm for Multiple Network Alignment

⇒ aims to optimize the # of conserved edges (MCES)

⇒ while also identifying partially conserved edges

Input: A set of networks X:

$X = \{G_1((V_1, E_1), \ldots, G_n(V_n, E_n)\}$, $|V_i| \leq N_{max}$, $\forall i \leq n$

Multiple Alignment, a set of mappings:

$A = \{f_i : V_1 \to V_i \mid 1 \leq i \leq n\}$

Goal: Find the alignment A that maximizes # fully conserved edges

$$f(A) = \sum_{(x,y) \in U_1} \begin{cases} 1, & \text{if } (f_i(x), f_i(y)) \in E_i, \forall i \in [1,\mu] \\ 0, & \text{else} \end{cases}$$

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CytoHCS

In: X, a set of networks

Out: A^*, alignment

$A \leftarrow$ Initial Alignment (X)
$A^* \leftarrow A$
repeat
  $A \leftarrow$ Perturb (A, X)
  $A \leftarrow$ Local Search (A, X)
  if $f(A) > f(A^*)$ then
    $A^* \leftarrow A$
  end
until not improving
return A^*

// Perturbation: avoids stuck in local maximum
Make random changes to current solution. Creates new starting point for local search

// Local Search:
Apply local changes to solution until no improvement can be made

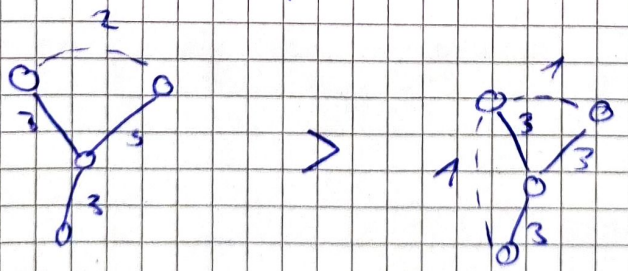
Partially conserved edges

→ local search maximizes the following fitness function:

$$g(A) = \sum_{(u,v)} C_A(u,v)^2$$

$$C_A(u,v) = \sum_{i=1}^{n} \begin{cases} 1, & \text{if } (f_i(u), f_i(v)) \in E_i \\ 0, & \text{else} \end{cases}$$

↳ Idea: Prioritize fewer well-conserved edges over many poorly conserved ones



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Local Search

In: A, alignment ; X, weights

Out: A', alignment

```
A' ← A
repeat
    for G_i in X, 2 ≤ i ≤ n do
        //Find swap that produces greatest increase in q(A')
        (u,v) ← argmax_{u,v ∈ V_i} delta(A', u, v)
        if delta(A', u, v) > 0 do
            A' ← swap(A', u, v)
        end
    end
until no improvement
return A'
```

- Local search procedure:
  Repeatedly swap the alignment of pairs to improve $q(A)$ until convergence
- Delta function: $delta(A, u, v) = q(swap(A, u, v)) - q(A)$
- Perturbation Procedure
  → randomly swaps 20% of $\forall$ pairs
  ↳ escaping local maxima :)

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GEDEVO

Evolutionary Graph Edit Distance Algorithm for Biological Network Alignment

→ Alignment between two networks that minimizes the graph edit distance (GED) wrt edges

→ Transform graph G1 into G2 with insdels

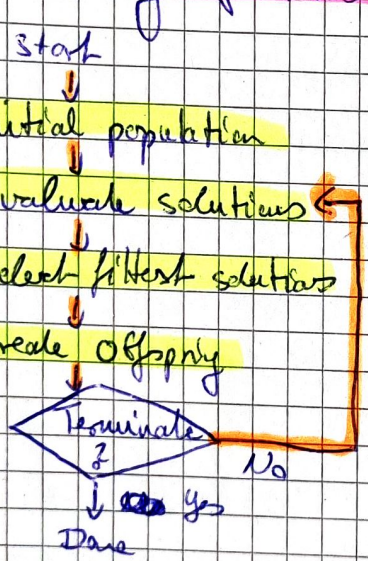
→ Fitness function

# of edges we need to insert/delete to make G1 isomorphic to G2

$$GED(f) = |\{(u,v) \in E_1 : (f(u), f(v)) \notin E_2\}| + |\{(u',v') \in E_2 : (f^{-1}(u'), f^{-1}(v')) \notin E_1\}|$$

→ use evolutionary algorithm to find a good solution

Evolutionary Algorithms



→ subset of evolutionary computation
→ maintains a population of candidate solutions
→ New solutions are created through mutation and crossover operations
→ only the fittest solution survive each generation

GSB

GREEDEUD

→ Initial population generation:
- → created from random permutations
- → reduce convergence time: Pre-align based on sequence similarity (BLAST)
  Pre-align based on local network topology

→ Evaluating node pairs:
- → use pairScore for local optimization when building offspring
  $u_1 \in V_1, v \in V_2, u = f(v)$

$$pairScore(u,v) = \frac{pairGED_f(u,v) + GDSD(u,v)}{2}$$

- → pairGED: relative number of induced edges incident to u or v when mapping u to v
- → GDSD: graphlet degree signature distance.
  A topological measure for how dissimilar the local neighborhoods of two nodes are

→ Graphlet: small, connected, non-isomorphic, induced subgraphs

→ Graphlet Degree Signature Distance, a generalization of node degree
  i.e. graphlet degree d_i^v is # of times node v touches a graphlet at orbit i

→ GDSD: how similar the signatures of two nodes are esp the same over all orbits

$$D_i(u,v) = w_i \times \frac{|\log(d_i^u + 1) - \log(d_i^v + 1)|}{\log(\max(d_i^u, d_i^v) + 2)}$$

$$GDSD(u,v) = \frac{\sum_i D_i(u,v)}{\sum_i w_i}$$

w_i weights are used to adjust for redundancy due to some orbits being part of others

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Offspring Generation

Strategies:

1) Random Generation: random permutation
2) Partially-mapped crossover mutation:
   Splits mapping into good and bad pairs and randomly permutes all bad pairs. Also permutes 1% of good pairs
3) Directed Mutations:
   Swaps a random number of randomly chosen bad pairs. Keep only the one that induces the best score
4) Crossover:
   Combines two or more existing solutions into a new one: Sorts pairs by score and adds pair from best to worst. Assigns remaining nodes randomly

Selectly surviving solutions

→ Solutions are initialized with health $H_i(0) = Q$; $Q > 0$

→ Fitness is computed by ranking solutions (lower is better)

$$C = \{ GED(f_i) : 1 \leq i \leq n \}, \quad r_i = \frac{rank(C_i)}{n}$$

→ the bottom 90% solutions lose health proportional to their rank:

$$H_i(t+1) = \begin{cases} H_i(t) - (r_i - 0,1) \times 0,9 \times \alpha & , \text{if } r_i > 0,4 \\ H_i(t) & , \text{else} \end{cases}$$

↳ if health ≤ 0 → remove this solution

Health Values

=> increases population diversity and helps escape local optimum

Termination

Possible:
- → after a N iterations
- → after T minutes
- → when the fitness of the best solution did not improve for

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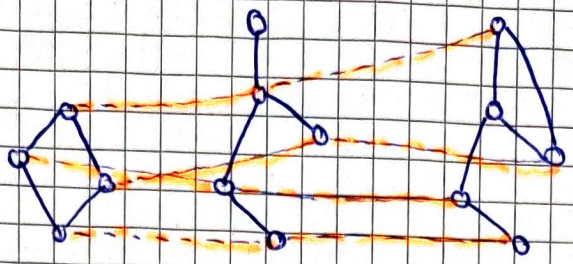
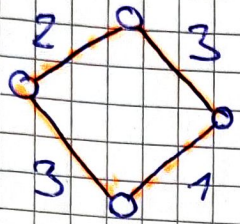
① CytoMCS

a) CytoMCS performs multiple, global network alignment

b) g(A)

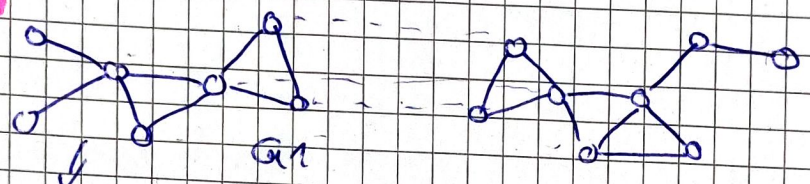




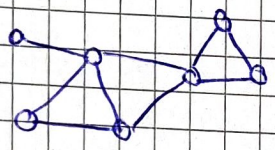
$$g(A) = 3^2 + 3^2 + 4\,2^2 + 1 = 23$$

② GEDEVO

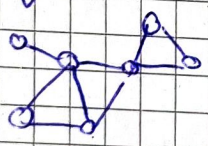
a)



inserted Edge

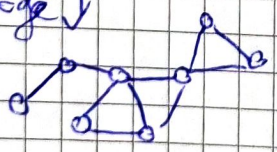


inserted Node



$$\Rightarrow GED(G1, G2) = 3$$

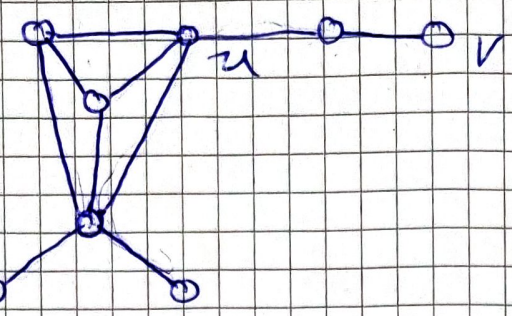
deleted Edge

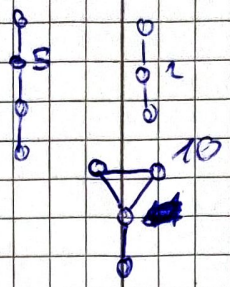


C S B

2)

b)





	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14
d(u)	4	3	3	3	0	5	1	0	0	0	7	3	0	0	1
d(v)	1	1	0	0	3	0	0	0	0	0	0	0	0	0	0

c) GDSD , w

$$D_i(u,v) = w_i \cdot \frac{|\log(d_i^u + 1) - \log(d_i^v + 1)|}{\log(\max(d_i^u, d_i^v) + 2)}$$

$$GDSD(u,v) = \frac{\sum_i D_i(u,v)}{\sum_i w_i}$$

d) two groups $V1 = \{A,B,C,D,E\}$, $V2 = \{F,G,H,I\}$, crossover procedure:

A	B	C	D	E
0,75	0,10	0,60	0,15	1,0
F	H	I	G	Ø

A	B	C	D	Ø	E
0,35	0,45	0,30	1	1	1
I	F	H	Ø	G	Ø

A	B	C	D	E
0,8	1	0,6	0,05	0,95
H	Ø	I	F	G

GOOD ← → BAD

D	B	D	C	A	B	C	A	A	E
0,05	0,1	0,25	0,3	0,35	0,45	0,6	0,75	0,80	0,95
F	H	G	H	I	F	I	F	H	G

E	D	Ø	E	B
1	1	1	1	1
Ø	Ø	G	Ø	Ø

No C here => Ø

=> Merge From left

A	B	C	D	E
I	H	Ø	F	G